**CU6051NI Artificial Intelligence**

**Project Title: Credit Card default Prediction**

**25% Individual Coursework**

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**Submitted To:** Mr. Alish KC

| | |
|---|---|
| **GitHub Link** | https://github.com/SabinAdhikarii/Credit-Card-Default-Detection-.git |

*I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.*

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# 1. Introduction

## 1.1 Introduction To The Project Title

In the digital age of todays, credit card default prediction is a serious issue especially in the financial industry where banks and other finance lending institutions must assess the risk of customer failing to meet repayment obligation. Credit card default prediction is a vital area in monetary and financial risk management. The increase in digitalization of have directly caused in growth of usage of credit card usage worldwide, financial institutions are facing increasing challenges in identifying customers who may fail to repay their obligations. Nepal not being very far, have also started digitalization in cards distributing credit card for first time in 1990 (the HRM, 2023). Since the day, Nepal also has distributed almost more than 318,428 credit cards in Nepal (INVESTOPAPER, 2025). The credit cards have been in an easy access to the people nowadays but it’s actually very hard to interpret if the user will be able to pay back the used credit amount or not.

For the solution of this problem, Artificial Intelligence (AI) can be used. This coursework allows us to make an AI system that can predict if the user can actually pay back the credit amount or not. For this a model will be trained using the past records of the user with various factors and predict defaults, enabling proactive risk mitigation.

## 1.2 Problem Domain

The Banks or any other financial institution that provide credit card and allows user to use credit up to some extents are facing huge losses and backlashes due to poor identification of the person who is likely to pay or not pay the credit amount. Traditional credit scoring system often rely on limited demographic or historical repayment data, which fails to capture the complex behavior and financial pattern that could influence default risk.

The imbalanced and inconsistent records where the number of non-default cases far outweighs default cases are also the challenge of this coursework. These imbalances made it harder for conventional models to detect rare but critical default events, leading to poor recall and precision of the model.

## 1.3 Project as a Solution

The proposed project is “Credit Card Default Detection System”, for solving the chances of frauds by early detection of the customer who is likely to be at high risk. The system will have a model that will be trained on a transaction data of the bank specifically of the credit card payment history, billing data, nature and behavior of the client. The model will analyze and dataset, finding the clients with high chance of risk. The system will allow back to identify the risky clients early and take any necessary actions, lower the credit limits, or add a friendly reminder for payments. The AI system will leverage the banks efficiency reducing both risk and loss amounts and successfully gaining more profits gradually.

## 1.4 Current Scenario and Issues/Challenges

The credit card default can be refereed as using or taking the benefits from the credit card using it more the purposes like shopping, paying, and getting exclusive offers but not paying at time or never paying back to the back. This problem exists both in global scale and at local levels. Global challenges mainly focused on regulatory pressure on banks to achieve a stability in finance while local challenges focusses on identifying the credit worthy people to achieve their financial stability.

Table 1: Comparison of Global Challenges & Local Challenges

| Global Challenges | Local Challenges (Nepal Context) |
|---|---|
| Increasing in fraudulent activities and defaults leading to billions in losses annually. | Increasing in debt of a person due to lack of awareness about repayment obligation. |
| Increasing reliance on credit cards for online transactions. | Increasing reliance on credit card for instant payment for goods or services. |
| Regulatory pressure on banks to maintain financial stability. | Banks face challenges in assessing credit worthiness due to incomplete financial histories. |
| Fraudulent activities and defaults leading to billions in losses annually. | Fraudulent activities as person unable to repay the amount to bank in time. |

## 1.5 AI Concepts Used In The Coursework

The project leverages several core AI concepts, the model will be first loaded and trained on labeled data as supervised learning methods. The classification of the data will be done using the classification algorithms like Logistic Regression, Random Forest, Gradient Boosting or others. The concept of feature engineering will be used for transforming raw demographic and financial data into meaningful inputs. For the cases like imbalanced or inconsistent data, data cleaning and handling techniques will be used and at last the model will be evaluated on using evaluation metrics if the prediction is satisfying the system is deployed otherwise the parameters are changed and evaluated again. The central goal is to use historical client data to predict whether a customer will default on their credit card payment in the following months. This is a classic binary classification problem, where target variable has two possible outcomes: default (1) and non-default (0).

### 1.5.1 Supervised Machine Learning

The dataset contains labeled data meaning model will learn from the labelled data, which makes it supervised machine learning. The algorithms being used here learns the pattern of the data from labelled data where the input features could be repayment history, credit limit, bill amounts, etc and the target variable is default or not.

### 1.5.2 Logistic Regression

Logistic regression is a linear model that estimates the probability of default using logistic function or sigmoid function. Here in this project, it is serving as a baseline model due to its simplicity and interpretability. It shows how demographic and financial variables relate to default risk.

### 1.5.3 Ensemble Methods

Ensemble Methods have been implemented in the coursework as two algorithms that uses ensemble method is used. Random Forest and Gradient Boosting both uses ensemble techniques. Random forest builds multiple decision tree and average their prediction reducing variance while Gradient boosting sequentially build tree where each new tree corrects the errors of the previous one. It is effective model for the coursework like credit card default (Github, 2025).

### 1.5.4 Model Evaluation Metrics

The model used in training and prediction are evaluated on few matrices. These matrices actually determine how the model is performing. While predicting a single output, the model checks it in many different parameters like accuracy, precision, recall f1 score and ROC-AUC.

**Accuracy:** Proportion of correctly predicted instances out of the total predictions made.

**Precision:** Proportion of correctly predicted positive cases (true positives) out of all cases predicted as positive (true positives + false positives).

**Recall:** Proportion of correctly predicted positive cases out of all true cases.

**F1:** Combination of precision and recall

**ROC-AUC:** ROC curve is used to visualize the trade-off between sensitivity and specificity, while AUC quantifies overall discriminative ability.

### 1.5.5 Feature Importance

Feature importance is one of the significant step to check after model prediction as it shows us which feature influenced most for the prediction. In the system like credit card default prediction, knowing which feature contributed more to the model or influence most helps businesses and financial institution to understand the credit risk factor. It helps identifying which features most influence predictions enhances interpretability. For the models like random forest and gradient boosting, feature importance scores highlight variables and default risk. This supports explainable AI (XAI) making model's decision more transparent for business or financial institution (SPRINGER NATURE Link, 2025).

## 1.6 Aim and Objectives

### 1.6.1 Aims

The aim of the project is to develop a supervised learning solution for predicting credit card defaults using AI model.

### 1.6.2 Objectives

- To collect and preprocess financial datasets.
- To train the AI models and test them.
- To evaluate performance using precision, recall, and F1-score, etc.
- To provide conceptual diagrams and pseudocode for implementation.

## 1.7 Dataset Selection for the Project

**Name**: Default of Credit Card Clients (Taiwan)

**Source**: UCI Machine Learning Repository

**Size**: 30,000 rows

**Features**: 24 variables

The dataset for the project is collected from the official website of UCI Machine Learning Repository. The dataset turn out to have more than 30,000 rows which is actually a great dataset for the real world projects like credit card defaults.

# 2. Background

## 2.1 Project Overview and Research Work Done

The project is titled as "Credit Card Default Detection". This project is an essential project required in the domain of finance to detect the default of credit so that the clients who is the red zone or in the verge where they might not be able to repay the obligations can be identified and necessary actions like freezing the current credit status, limiting the credit amount, stopping the payments or more can be done. The project focusses on implementing supervised learning as the dataset, itself provides labeled outcome as default or non-default. This approach is the standard industry align practices which even offer measurable accuracy improvement. The system tends to implement at least of three algorithms. They are: Logistic Regression, Decision Tree and Random Forest. The system will evaluate the recall, precision and F1 score of all the models and evaluate the model with the highest accuracy. The performance of any classification algorithm can be influenced by the dataset. Hence, all three algorithm's output will be analyzed comparatively.

The justification for this project lies in its real-world applicability:

- It helps banks and other financial institution to reduce their debts and losses by identifying high-risk clients early.
- It supports regulators in maintaining financial stability.
- It benefits customers by ensuring fairer lending policies and proactive support.

## 2.2 Research and Review of Similar Projects

Credit card default prediction is a serious problem in the financial domain. Banks and financial institutions face significant risk when client fails to repay their credit amount. Before the development of our project, it is important to do research if the project or complete or partially have already been done by other people also. During my research to check if any other similar project is already available or not. I found some of the similar projects. The project is done on similar basis but not the exact project had done. With different virtue of the people, the inspiration to conduct the project more effectively and efficiently can be clearly seen from the research paper and similar projects. Among the six seven similar projects and research work I have seen most people have used machine learning algorithms where some have used the deep learning techniques, some did the project using neural networking and all but the outcome of the overall project was always the same. The output given by the project is almost similar. The ground for the development of the project were same for default credit card detection but they have used different dataset, different algorithms and techniques. I have shortlisted three similar projects. They are:

### 2.2.1 Consumer Credit-Risk Models via Machine-Learning Algorithms

"Consumer Credit Risk Models via Machine Learning Algorithms" is the one of the similar projects done by Amir E. Khandani, Adlar J. Kim, and Andrew W. Lo. The project mainly focuses on implementing machine learning to big financial data for customer credit risk. The tree-based algorithm (CART-like) is used to build models and compared with traditional methods (HAL-SH, 2010). The authors compared the performance of these machine learning models against credit scoring method and demonstrated the machine learning superior predictive accuracy and robustness. This project by Khandani, Kin, and Lo(2010) is similar to our project as both project uses machine learning models to find the accuracy of the credit default. Also, both projects are emphasizing the importance of behavioral and finance features in building robust models.

**Algorithm Used:** The developers of consumer credit risk model project have used (CART-like models) to construct nonlinear, nonparametric forecasting models of consumer credit threat or risk. These models were compared with traditional credit scoring methods like linear regression, and even with logistic regression. The algorithm used allowed them to capture the essence of behavioral and financial feature which is often missed or failed by traditional parametric models.

**Dataset Used:** The dataset used in this project is the combination of customer transaction records and credit bureau data from a major commercial bank in the United States of America. The dataset have covered the time period of 4 years ranging from 2005 to 2009. The dataset was large, lengthy including the demographic information, credit card usage repayment behavior and delinquency history (MIT Libraries, 2010).

**Outcome of the project**: The research and project study found out that machine learning models significantly outperformed traditional credit scoring methods in predicting delinquencies and defaults. The achievement of the project was it out smartered the traditional approach also out of sample forecast achieved classification rates with r-square values of up to 85% between forecasted and realized delinquencies. The model was robust, and predicted the data with high accuracy even under conservative assumptions.

### 2.2.2 Random Forest for peer-to-peer lending defaults.

Another important project in the field of credit default detection which is also similar to our project was conducted by Malekipirbazari and Aksakalli in 2015 titled as 1. Random Forest for peer-to-peer lending defaults. This study was more aligned toward growing peer-to-peer lending platforms, where individual lend money directly to the other individuals without traditional banking intermediaries (Y. N. Ifriza, 2015). The author recognized default risk in peer-to-peer and applied Random Forest algorithm to ensemble it with decision tree to improve the accuracy and reduces the overfitting. This project concludes that Random Forest is a great choice for the project like credit risk detection (Aksakalli, 2018).

**Algorithm Used:** The study done by Malekipirbazari and Aksakalli in 2015 uses Random Forest algorithm. The algorithm uses ensemble learning method that builds multiple decision trees and later aggregate their prediction and combining their prediction it provide single prediction which is likely to get higher. The reason for chosing random forest by the author is because it reduces the overfitting in comparison to single decision tree and provide high accuracy(Y. N. Ifriza, 2015).

**Dataset Used:** The dataset for the project is collected from **LendingClub,** it is one of the largest peer to peer lending platforms in the United States of America. It have the informations on loan application borrower demographics, loan amounts, interest rates, and repayment outcomes. The dataset contains thousabds of loan recods making it a rich source for analyzing default risk in peer to peer lending markets.

**Outcome of the work:** The study shows that the Random Forest Classifier significantly outperformed traditional credit scoring methods in predicting defaults on peer to peer loans. The key findings of this projects was that random forest is a better option if we are doing the classification on similar data other than logistic regression and other baseline models. The model also highlighted the importance of borrower financial attribute in default prediction.

### 2.2.3 Credit Risk Prediction Using Machine Learning and Deep Learning

This project serves the similar project operations as it compares the several algorithms like Logistic regression, AdaBoosr, XGBoost, LightGBM, and Neural Networks so that the model could predict the best accurate ouput (Chang, 2024). This research was done by Victor Chang, Sharuga Sivakulasingam, Hai Wang, Siu Tung Wong, Meghana Ashok Ganatra, and Jiabin Luo in 2024. The project is similar to ours as it focuses on credit card default prediction, uses supervised learning and perform ensemble methods.

**Algorithm Used:** The algorithm used by Victor Chang, Sharuga Sivakulasingam, Hai Wang, Siu Tung Wong, Meghana Ashok Ganatra, and Jiabin Luo in 2024 implemented a wide range of mchine learning and deep learning to predict credit default. The algorithms included Logistic Regression, AdaBoost, XGBoost, LightGBM, and Neural Networks. The ensemble methods (AdaBoost, XGBoost, LightGBM) were particularly emphasized for their ability to handle complex feature interactions and imbalanced dataset.

**Dataset Used:** The dataset used in this research was a large scale credit card default dataset is also taken from UCI Machine Learning Repository. The dataset to use in this study consisted of a huge credit card default sample that was similar to that of Taiwan UCI however this data was supplemented with some additional financial and behavioral data. It consisted of tens of thousands of records with demographic, financial and repayment history characteristics. The choice of this data set was to represent the realistic patterns of using credit card thus making the study more relevant to financial institutions seeking to come up with effective predictive models.

**Outcome of the work:** It was established in the project that the ensemble learning model especially the XGBoost and LightGBM, produced the greatest predictive power when compared with traditional models like the Logistic Regression. Neural networks also showed good results but required a higher computing power and a careful adjustment of the hyperparameters. The main conclusions are as follows:

- XGBoost and LightGBM outperformed other models with respect to accuracy, precision and recall.
- Logistic Regression was used as a control condition but was not as suitable in complex relationships capture.
- -Neural networks were sufficiently promising but had a lack of interpretability with respect to tree-based ensembles.

The paper has highlighted that integrative method wise, machine learning approach coupled with deep learning approach offers a well-rounded trade- off between accuracy, robustness and interpretability.

## 2.3 Review and Comparison of Similar Projects

Table 2: Comparison Between Similar Project

| Project | Algorithm | Dataset | Accuracy | Strengths | Weaknesses |
|---|---|---|---|---|---|
| Khandani et al. (2010) | ML models (CART-like trees) | US consumer credit data | ~80% | Rich dataset, captures behavioral patterns | Complex preprocessing, computationally heavy |
| Malekipirbazari & Aksakalli (2015) | Random Forest | Peer-to-peer lending platform | ~85% | High accuracy, robust to imbalance/noise | Interpretation is low. |
| Victor Chang et al. (2024) | ML & Deep Learning (Neural Networks) | Credit card customer dataset | ~85–90% | Ensemble + deep learning superior recall and ROC- AUC | Requires high computational resources, lower interpretability |

## 2.4 DATASET BACKGROUND

### 2.4.1 History of Dataset

The “Default of Credit Card Clients” dataset has its origins in 2009 study conducted by Cheng yeh and Che-hui Lien. They collected data from a Taiwanese financial institution. The motto of their project is to investigate the credit card risk assessment in Taiwan. The dataset only gets open for public after the accomplishment of their project which is published in Expert System with Application. The dataset initially contains more than 30,000 rows records with 23 input features only. The features consists of age, gender, education, and marital status, as well as financial attributes like credit limits, bill statements, and repayment history. The target variable, (name default payment next month) is a binary variable which represents defaulting of a client, thus making the data suitable to the supervised learning algorithms. The data set was subsequently gifted to the UCI Machine Learning Repository in January 2016, where it has since become one of the most widely-used resources in academic research and education in the field of credit-risk modeling. Its historical importance can be attributed to its realistic modeling of real-world financial behavior in Taiwan in the mid-2000s, and its lasting usefulness in the established use as a canonical reference point in the assessment of classification algorithms, ensemble methods, and also in investigating issues, like class imbalance and feature association.

### 2.4.2 Dataset Characteristics

| Attribute | Description |
|---|---|
| **Name** | Default of Credit Card Clients (Taiwan) |
| **Source** | UCI Machine Learning Repository (donated January 2016) |
| **Original Authors** | I-Cheng Yeh and Che-hui Lien (2009) |
| **Publication** | *Expert Systems with Applications* (2009), DOI: 10.1016/j.eswa.2007.12.020 |
| **Country of Origin** | Taiwan |

| | |
|---|---|
| **Instances (Rows)** | 30,000 credit card clients |
| **Features (Columns)** | 23 input variables + 1 target variable |
| **Target Variable** | default.payment.next.month (binary: 1 = default, 0 = no default) |
| **Feature Types** | Mix of categorical (e.g., gender, education, marriage) and numerical (e.g., credit limit, bill amounts, payment history) |
| **Era of Data** | Mid-2000s (dataset reflects financial behavior during 2005–2006) |
| **Class Distribution** | ~22% default, ~78% non-default (imbalanced dataset) |
| **Applications** | Credit risk modeling, supervised learning, classification, ensemble methods |

Table 3: Dataset Characteristics

# 3. Solution

## 3.1 Explanation of the Proposed Solution

The proposed solution to the problem of credit card default prediction is a development of a machine learning model that could analyze the customer or client's demographic data, repayment history and billing records to predict if a client is likely to default or mot. Here in this system supervised learning will be used meaning the model will learn from the labeled data sets to train to identify the underlying patterns and relationship (Ivan Belcic, 2025). This solution helps banks and financial institutions to proactively identify high-risk clients, reduce financial losses and improve credit risk management system smoothly.

## 3.2 Explanation of AI algorithms Used

The project specifically focused on implementing three classification algorithms for the evaluation of the AI model developed. After allocating the training and testing data from in the ration 80:20, the algorithms are used to learn from the training and testing data, it usually recognizes patterns and make prediction (GeeksforGeeks, 2023). Since, our project majorly falls under identifying the likelihood of a person to check if he/she could actually repay the obligation or not, the classification technique is best. Under the classification technique, Logistic Regression, Random Forest and Gradient Boosting are the three algorithms being used in our project.

### 3.2.1 Logistic Regression

Logistic Regression: Logistic regression, though having regression in its name falls under classification. It is an algorithm under supervised machine learning used for classification problem. It is used for binary classification problems like True/False, Yes/No, in our case defaulter/non-defaulter. It works by concerting the inputs into probability values between 0 and 1 (GeeksforGeeks, 2025). In our project, it serves as a baseline model that interprets and provides insights into default risk. The Logistic Regression uses sigmoid function for classification and its value always ranges in between 0 and 1.

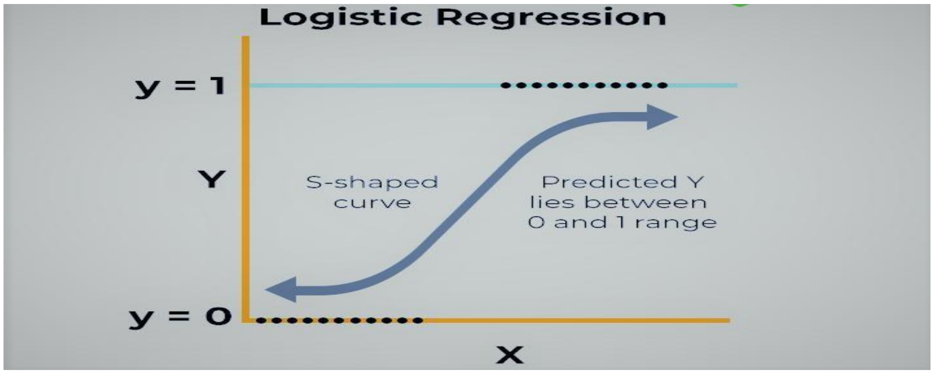



*Figure 1: Logistic Regression*

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

*Figure 2: Sigmoid Function*

### 3.2.2 Random Forest

Random Forest: Random Forest is an AI algorithm that uses ensemble techniques to make the prediction from decision tree. As the name suggests, it is the combination of large number of trees. It creates a large number of decision-tree trained on different dataset and combined all of them while making prediction (tutorialspoint, 2023).

In our project, this algorithm will be used for making prediction based on the decision made by multiple decision tree. The final prediction will be made combining the prediction of all decision tree.

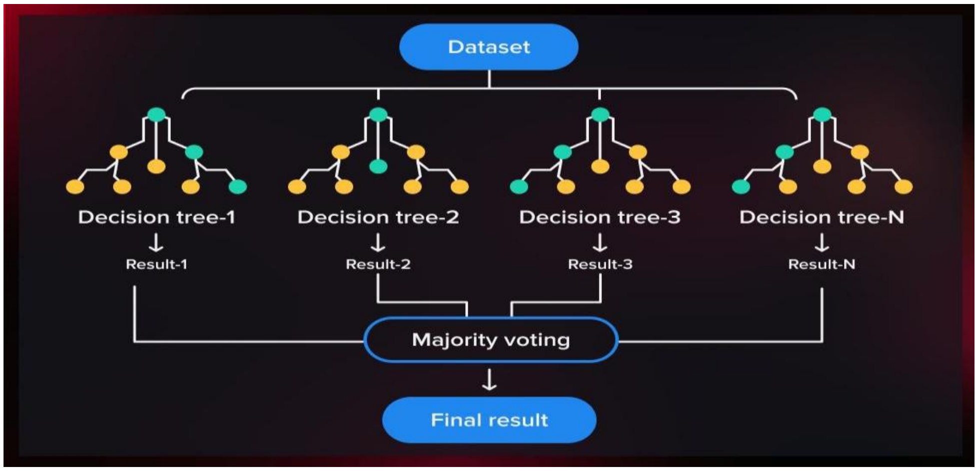



*Figure 3: Random Forest*

$$\hat{f} = \frac{1}{B} \sum_{b=1}^{B} f_b(x')$$

*Figure 4: Random Forest Classifier*

### 3.2.3 Gradient Boosting

Gradient Boosting: Gradient Boosting, a supervised machine learning algorithm which produces accurate and precise prediction by combining multiple decision trees into a single model (Bryah Clark, 2024). The algorithm work by combining the weak prediction into single ensemble. This algorithm minimizes the loss function and improves the predictive accuracy (Bryah Clark, 2024)

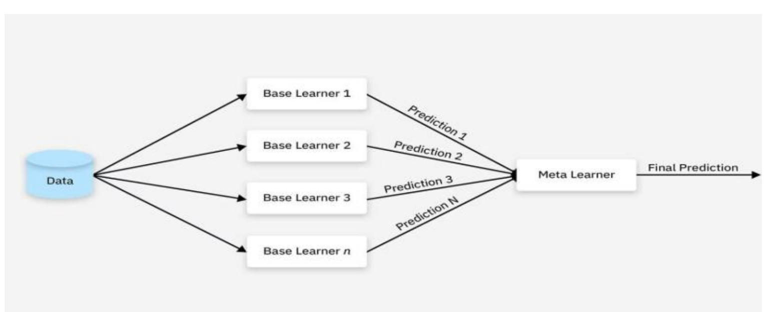



*Figure 5: Gradient Boosting*

$$F_m(x) = F_{m-1}(x) + \nu \cdot h_m(x)$$

Figure 6: Gradient Boosting formula

## 3.3 Diagrammatical Representation

### 3.3.1 System Flowchart

*Figure 7: System Flowchart*

### 3.3.2 System Workflow Diagram

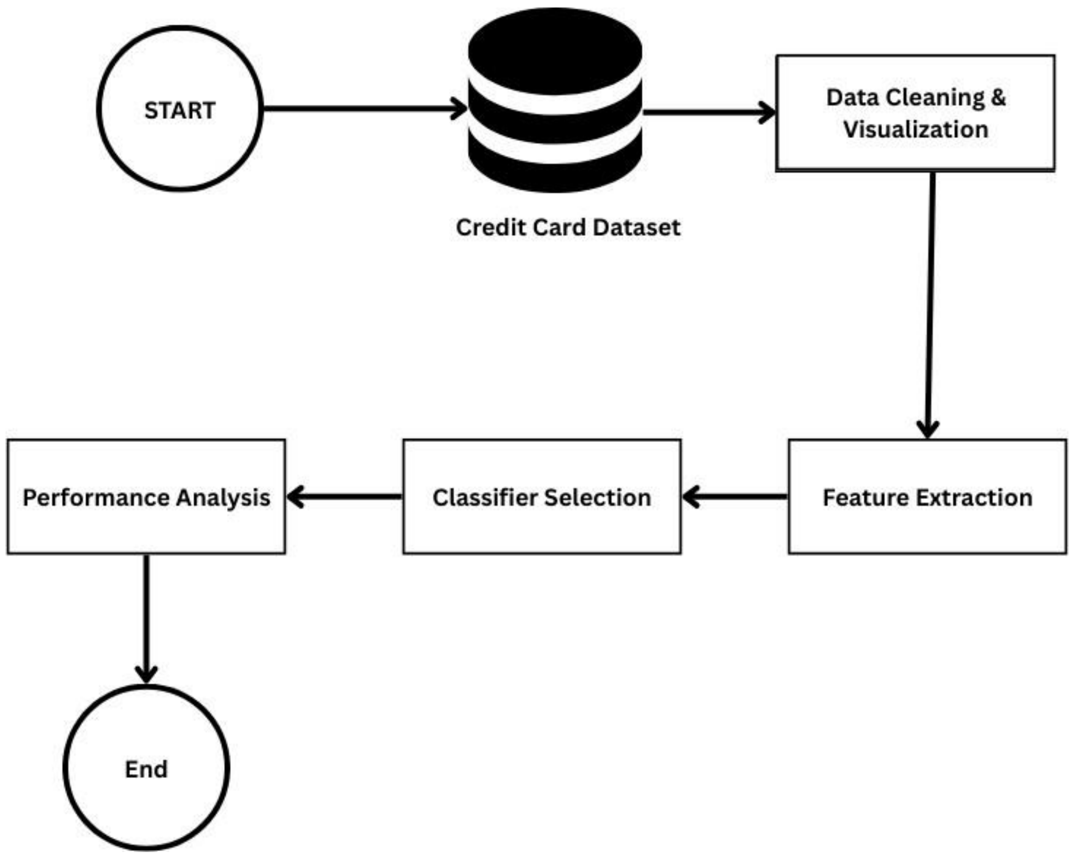



*Figure 8: System Workflow Diagram*

### 3.4 Pseudocode

```
BEGIN
    IMPORT libraries (pandas, numpy, sklearn, matplotlib, seaborn, imblearn)
    SET display_options for better visualization
    SET random_seed = 42 for reproducibility
    LOAD dataset FROM uploaded_file
    IF file_type == 'csv' THEN
        dataset = READ_CSV(filename)
    ELSE IF file_type IN ['xls', 'xlsx'] THEN
        dataset = READ_EXCEL(filename)
    END IF
    DISPLAY dataset.shape
    DISPLAY dataset.head()
    DISPLAY dataset.info()
    DISPLAY dataset.describe()
    IDENTIFY target_column FROM column_names WHERE 'default' IN column_name
    CREATE clean_dataset = COPY(dataset)
    REMOVE id_columns FROM clean_dataset
    FOR each column IN clean_dataset DO
        IF column HAS missing_values THEN
            IF column IS numeric THEN
                FILL missing_values WITH median(column)
            ELSE
                FILL missing_values WITH mode(column)
            END IF
        END IF
    END FOR
    REMOVE duplicate_rows FROM clean_dataset

    IF 'EDUCATION' IN columns THEN
```

```
    REPLACE values [0, 5, 6] WITH 4 IN EDUCATION
END IF

IF 'MARRIAGE' IN columns THEN
    REPLACE value 0 WITH 3 IN MARRIAGE
END IF

CREATE correlation_matrix FROM clean_dataset
VISUALIZE correlation_matrix AS heatmap

VISUALIZE target_distribution AS pie_chart
VISUALIZE target_distribution AS bar_chart
VISUALIZE age_distribution AS histogram
VISUALIZE credit_limit_distribution AS histogram
VISUALIZE gender_distribution AS bar_chart
VISUALIZE education_distribution AS bar_chart
VISUALIZE marriage_distribution AS bar_chart
VISUALIZE default_by_gender AS grouped_bar_chart
VISUALIZE default_by_education AS grouped_bar_chart

SEPARATE features_X FROM clean_dataset EXCLUDING target_column
SEPARATE target_y FROM clean_dataset AS target_column

CALCULATE class_imbalance_ratio = count(class_0) / count(class_1)

SPLIT features_X, target_y INTO:
    X_train (80%), X_test (20%)
    y_train (80%), y_test (20%)
    WITH stratification ON target_y
INITIALIZE scaler = StandardScaler()
```

```
    FIT scaler ON X_train
    X_train_scaled = TRANSFORM X_train USING scaler
    X_test_scaled = TRANSFORM X_test USING scaler

    IF class_imbalance_ratio > 2 THEN
        INITIALIZE smote = SMOTE(random_state=42)
        X_train_resampled, y_train_resampled = APPLY smote ON X_train_scaled, y_train
    ELSE
        X_train_resampled = X_train_scaled
        y_train_resampled = y_train
    END IF

    INITIALIZE models_dictionary = EMPTY
    INITIALIZE results_dictionary = EMPTY

    model_lr = LogisticRegression(max_iter=1000, random_state=42)
    FIT model_lr ON X_train_resampled, y_train_resampled
    STORE model_lr IN models_dictionary['Logistic Regression']

    model_rf = RandomForestClassifier(n_estimators=100, max_depth=10,
random_state=42)
    FIT model_rf ON X_train_resampled, y_train_resampled
    STORE model_rf IN models_dictionary['Random Forest']

    model_gb = GradientBoostingClassifier(n_estimators=100, learning_rate=0.1,
max_depth=5, random_state=42)
    FIT model_gb ON X_train_resampled, y_train_resampled
    STORE model_gb IN models_dictionary['Gradient Boosting']

    FOR each model_name, model IN models_dictionary DO
```

```
    y_pred = PREDICT model ON X_test_scaled
    y_pred_proba = PREDICT_PROBABILITY model ON X_test_scaled

    accuracy = CALCULATE_ACCURACY(y_test, y_pred)
    precision = CALCULATE_PRECISION(y_test, y_pred)
    recall = CALCULATE_RECALL(y_test, y_pred)
    f1_score = CALCULATE_F1_SCORE(y_test, y_pred)
    roc_auc = CALCULATE_ROC_AUC(y_test, y_pred_proba)

    confusion_matrix = CALCULATE_CONFUSION_MATRIX(y_test, y_pred)

    STORE {
        accuracy, precision, recall, f1_score, roc_auc,
        confusion_matrix, y_pred, y_pred_proba
    } IN results_dictionary[model_name]

    DISPLAY model_name
    DISPLAY accuracy, precision, recall, f1_score, roc_auc
    DISPLAY confusion_matrix
    DISPLAY classification_report

END FOR

CREATE comparison_dataframe FROM results_dictionary WITH columns:
    [Model, Accuracy, Precision, Recall, F1-Score, ROC-AUC]

DISPLAY comparison_dataframe
```

```
    best_model_name = FIND model WITH MAXIMUM f1_score FROM
comparison_dataframe
    best_model = GET models_dictionary[best_model_name]
    best_results = GET results_dictionary[best_model_name]

    DISPLAY "BEST MODEL: " + best_model_name
    DISPLAY best_results

    VISUALIZE metrics_comparison AS grouped_bar_chart

    FOR each model IN results_dictionary DO
        VISUALIZE confusion_matrix AS heatmap
    END FOR

    FOR each model IN results_dictionary DO
        CALCULATE fpr, tpr FROM roc_curve(y_test, y_pred_proba)
        PLOT roc_curve(fpr, tpr) WITH label = model_name + AUC_score
    END FOR

    IF model_name == 'Random Forest' THEN
        feature_importances = GET model.feature_importances_
        SORT feature_importances IN descending_order
        SELECT top_10_features
        VISUALIZE top_10_features AS horizontal_bar_chart
    END IF

    EXTRACT tn, fp, fn, tp FROM best_results.confusion_matrix

    DISPLAY detailed_analysis:
        True_Negatives = tn
```

```
        False_Positives = fp
        False_Negatives = fn
        True_Positives = tp

    SELECT random_sample OF size 10 FROM X_test

    X_sample_scaled = TRANSFORM random_sample USING scaler
    y_sample_true = GET actual_labels FOR random_sample

    y_sample_pred = PREDICT best_model ON X_sample_scaled
    y_sample_proba = PREDICT_PROBABILITY best_model ON X_sample_scaled

    CREATE prediction_dataframe WITH columns:
        [Actual, Predicted, Default_Probability, Correct]

    DISPLAY prediction_dataframe
    SAVE comparison_dataframe TO 'model_comparison_results.csv'
    SAVE best_model TO 'best_model_*.pkl' USING pickle
    SAVE scaler TO 'scaler.pkl' USING pickle

    DISPLAY project_summary:
        Total_Samples
        Number_of_Features
        Target_Variable
        Class_Distribution
        Models_Trained
        Best_Model_Name
        Best_Model_Metrics
        Saved_Files
END
```

## 3.5 Tools and Technology Used

For the development of the project, different tools and technologies had been used. As the project mostly lean toward the machine learning, the machine learning algorithm is used along with the its libraries. Google collab is used as the platform or IDE for the development of the project. The project is solely developed in the Google collab. The tech stack used is python as python is AI and ML friendly language with many pre-defined libraries.

Some of the tools and tech stack used are mentioned below:

### 3.5.1 Python Programming Language

Python is a high-level, general purpose programming language renowned for its simplicity and readability. For the development of any AI and Machine Learning project, Python is must. Python already have thousands of inbuilt libraries that can handle and support the machine learning and artificial intelligence. Python has been acting as a backbone of artificial intelligence.

*Figure 9: Python Programming Language*

### 3.5.2 Pandas and Numpy for Data Handling

Pandas and Numpy both are the libraries of pandas used in data science, machine learning and artificial intelligence for manipulating and handling data. Pandas is used for manipulating the dataset, for e.g loading the datasets, checking the columns, adding or deleting columns while numpy can add two or more columns numeric data. It can convert the date time into different formats. Numpy is one of the most important libraries of pandas used.

*Figure 10: Pandas and Numpy Library*

### 3.5.3 Sci-kit Learn

Sci-kit Learn also known as sklearn is a open source library of machine learning in python which built on Numpy, SciPy, and matplotlib. It acts as an API provider for a variety of supervised machine learning and unsupervised machine learning algorithms for the crucial steps like data processing, model selection, model training and model evaluation. It supports the major task like classification, regression, clustering, dimensionality reduction and also feature engineering (Sci-kit Learn , 2024).

*Figure 11: Sci-kit Learn*

### 3.5.4 Matplotlib

Matplotlib is an open-source visualization library for the python programming language, widely used for making graphs, charts and other visualization models like heatmap, bar graph, pie chart, etc. Matplotlib is a technique that allows user to better understand data and model (GeeksforGeeks, 2025).

Matplotlib allows to generate both 3D visualizations and 2D visualization.

*Figure 12: Matplotlib*

### 3.3.5 Google Colab

Google colab is a jupyter notebook service hosted online meaning it is a cloud based environment that allows user to computer resources, write and execute python codes through their web browser. The google colab has inbuilt GPUs and TPUs that let developer train the model much faster than on CPU. Here in our project, Google Colab has been used as an IDE for the development of the project.

*Figure 13: Google Colab*

# 4. Explanation of the Development Process

The development of this project followed a structured machine learning pipeline to ensure reproducibility, and interpretability. The development process began with the research work started about the credit default. During the research many key identity and attributes of the credit default system is needed. After knowing about the feasibility test, the next step included dataset. A detailed dataset was needed, we get it from UCI Machine Learning Repository. The dataset is then loaded to the system. After loading the dataset, the step involved is data exploration and preprocessing. Explanatory Data Analysis (EDA) was conducted to understand the distribution of the target variable, identify class imbalance and examine the statistical properties of the features. Missing values were checked and numerical features were standardized using sci-kit learn's Standard Scale. After that the dataset is split into train test with the ratio of 80/20.

The models were declared and data is fed, after the completion of training the models were evaluated. For the model performance evaluation, ROC Curve and confusion matrices are created.

## 4.1 Installing Required Libraries

The project was developed in Google Colab which already have the necessary libraries installed. However, for the people running in different IDE may need to install them.

Step 1: Installing and Import Required Libraries

```
#Installing Required packages
!pip install numpy
!pip install pandas
!pip install matplotlib
!pip install seaborn
!pip install scikit-learn
!pip install warnings
```

*Figure 14: Screenshot of Installing Required Libraries*

## 4.2 Importing Required Libraries

The second stage is to import all the required packages for the operations. Without importing these packages non any of the functions will work.

```
[ ]
#Importing all the required libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifier
from sklearn.metrics import (accuracy_score, precision_score, recall_score,
                             f1_score, roc_auc_score, confusion_matrix,
                             classification_report, roc_curve)
import warnings
warnings.filterwarnings('ignore')

sns.set_style('whitegrid')
plt.rcParams['figure.figsize'] = (10, 6)

print("All libraries imported successfully!")

All libraries imported successfully!
```

*Figure 15: Screenshot of Importing Required Libraries*

## 4.3 Loading Dataset

The dataset from UCI Repository is downloaded and loaded manually. For loading the dataset pandas library is used. As the dataset was in excel format, pandas is used with excel for data handling and loading.

Step 2: Load the Dataset

```
[5]
1s
print(" LOADING DATASET...")

df = pd.read_excel('/content/Credit-Card-Default-Dataset(Taiwan).xls')
df.head(10)

LOADING DATASET...
```

| | Unnamed: 0 | X1 | X2 | X3 | X4 | X5 | X6 | X7 | X8 | X9 | ... | X15 | X16 | X17 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | ID | LIMIT_BAL | SEX | EDUCATION | MARRIAGE | AGE | PAY_0 | PAY_2 | PAY_3 | PAY_4 | ... | BILL_AMT4 | BILL_AMT5 | BILL_AMT6 | PAY_ |
| 1 | 1 | 20000 | 2 | 2 | 1 | 24 | 2 | 2 | -1 | -1 | ... | 0 | 0 | 0 | |
| 2 | 2 | 120000 | 2 | 2 | 2 | 26 | -1 | 2 | 0 | 0 | ... | 3272 | 3455 | 3261 | |
| 3 | 3 | 90000 | 2 | 2 | 2 | 34 | 0 | 0 | 0 | 0 | ... | 14331 | 14948 | 15549 | |
| 4 | 4 | 50000 | 2 | 2 | 1 | 37 | 0 | 0 | 0 | 0 | ... | 28314 | 28959 | 29547 | |
| 5 | 5 | 50000 | 1 | 2 | 1 | 57 | -1 | 0 | -1 | 0 | ... | 20940 | 19146 | 19131 | |
| 6 | 6 | 50000 | 1 | 1 | 2 | 37 | 0 | 0 | 0 | 0 | ... | 19394 | 19619 | 20024 | |
| 7 | 7 | 500000 | 1 | 1 | 2 | 29 | 0 | 0 | 0 | 0 | ... | 542653 | 483003 | 473944 | 5 |
| 8 | 8 | 100000 | 2 | 2 | 2 | 23 | 0 | -1 | -1 | 0 | ... | 221 | -159 | 567 | |

*Figure 16: Screenshot of Loading Dataset*

## 4.4 Data Exploration

Data Exploration includes check the overall statistics of the dataset, missing values, class distribution and so on. Here in our project, I have checked the shape, size, info, missing values and class distribution.

Step 3: Data Exploration

```
print(" DATA EXPLORATION")
df.info()
```

```
 DATA EXPLORATION
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30001 entries, 0 to 30000
Data columns (total 25 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Unnamed: 0   30001 non-null  object
 1   X1           30001 non-null  object
 2   X2           30001 non-null  object
 3   X3           30001 non-null  object
 4   X4           30001 non-null  object
 5   X5           30001 non-null  object
 6   X6           30001 non-null  object
 7   X7           30001 non-null  object
 8   X8           30001 non-null  object
 9   X9           30001 non-null  object
 10  X10          30001 non-null  object
 11  X11          30001 non-null  object
 12  X12          30001 non-null  object
 13  X13          30001 non-null  object
 14  X14          30001 non-null  object
 15  X15          30001 non-null  object
 16  X16          30001 non-null  object
 17  X17          30001 non-null  object
 18  X18          30001 non-null  object
```

*Figure 17: Screenshot of Check df info*

```
#Checking shape and size of dataset
print(f"\nDataset Shape: {df.shape}")
print(f"Total Records: {df.shape[0]:,}")
print(f"Total Features: {df.shape[1]}")
```

```
Dataset Shape: (30001, 25)
Total Records: 30,001
Total Features: 25
```

*Figure 18: Screenshot of Checking Shape and Size of Dataset*

```
#Checking for missing values
missing = df.isnull().sum()
print("Missing Values:")
print("No missing values!" if missing.sum() == 0 else missing[missing > 0])
```

```
Missing Values:
No missing values!
```

*Figure 19: Screenshot of Checking for Missing Values*

```
df.describe()  # description of dataset
```

| | Unnamed: 0 | X1 | X2 | X3 | X4 | X5 | X6 | X7 | X8 | X9 | ... | X15 | X16 | X17 | X18 | X19 | X20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| count | 30001 | 30001 | 30001 | 30001 | 30001 | 30001 | 30001 | 30001 | 30001 | 30001 | ... | 30001 | 30001 | 30001 | 30001 | 30001 | 30001 |
| unique | 30001 | 82 | 3 | 8 | 5 | 57 | 12 | 12 | 12 | 12 | ... | 21549 | 21011 | 20605 | 7944 | 7900 | 7519 |
| top | 30000 | 50000 | 2 | 2 | 2 | 29 | 0 | 0 | 0 | 0 | ... | 0 | 0 | 0 | 0 | 0 | 0 |
| freq | 1 | 3365 | 18112 | 14030 | 15964 | 1605 | 14737 | 15730 | 15764 | 16455 | ... | 3195 | 3506 | 4020 | 5249 | 5396 | 5968 |

4 rows × 25 columns

*Figure 20: Screenshot of Dataset Description*

```
# Check class distribution
print("\n\nTarget Variable Distribution:")
if 'default payment next month' in df.columns:
    target_col = 'default payment next month'
elif 'default.payment.next.month' in df.columns:
    target_col = 'default.payment.next.month'
else:
    # Find the last column (usually the target)
    target_col = df.columns[-1]

print(df[target_col].value_counts())
print(f"\nClass Distribution (%):")
print(df[target_col].value_counts(normalize=True) * 100)
```

```
Target Variable Distribution:
Y
0                              23364
1                               6636
default payment next month         1
Name: count, dtype: int64

Class Distribution (%):
Y
0                              77.877404
1                              22.119263
default payment next month      0.003333
Name: proportion, dtype: float64
```

*Figure 21: Screenshot of Checking for Class Distribution*

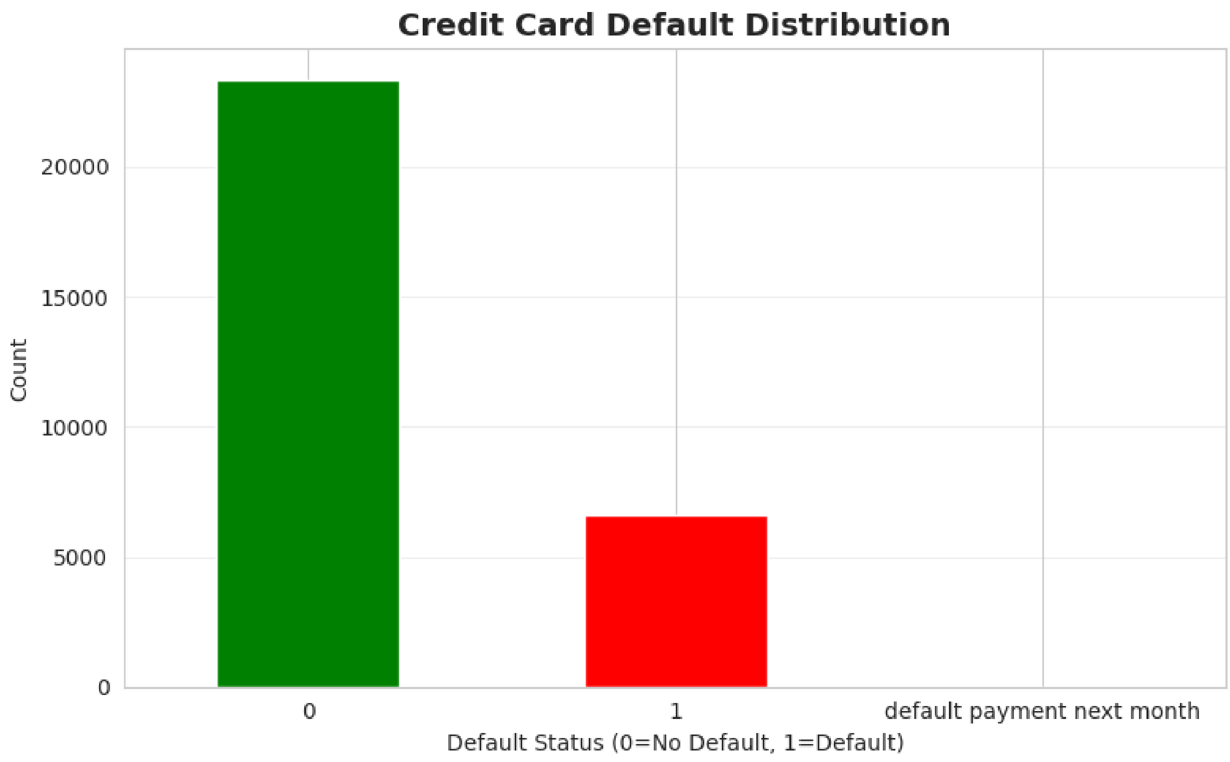



*Figure 22: Screenshot of Class Distribution Graph*

## 4.5 Data Preprocessing

The data preprocessing is a crucial part of the system development process. In this step the missing values are handled, imbalance classes are balanced. The columns are divided into features and targets, dataset is cleaned and processed for plotting. The numeric columns are standardize and train test data is divided.

## Step 4: Data Preprocessing

```
#Separating columns into feaure and targets
print("DATA PREPROCESSING")

X = df.drop([target_col, 'ID'], axis=1, errors='ignore')
y = df[target_col]

print(f"Features: {X.shape}")
print(f"Target: {y.shape}")
```

```
DATA PREPROCESSING
Features: (30001, 24)
Target: (30001,)
```

*Figure 23: Screenshot of Separating Features and Targets*

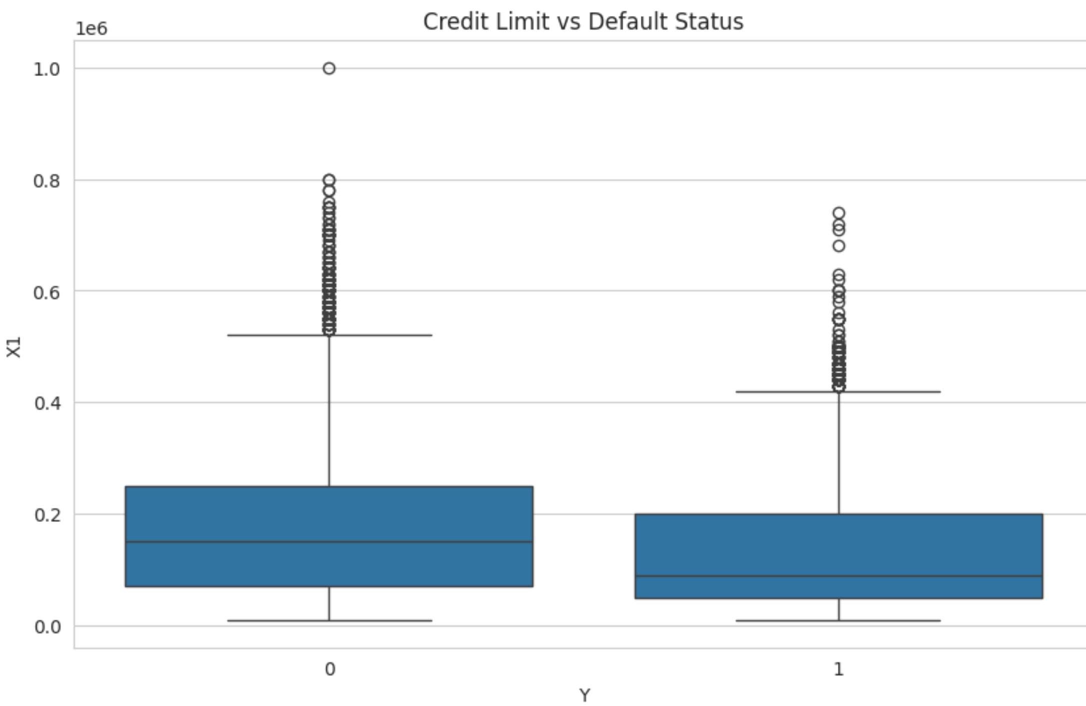



*Figure 24: Screenshot of Graph Separating Features and Targets*

```
# Scalig Features
print("Scaling features...")
X_temp = X.copy()
y_temp = y.copy()

X_temp = X_temp.iloc[1:]
y_temp = y_temp.iloc[1:]
X_temp = X_temp.apply(pd.to_numeric, errors='coerce')

# Convert y_temp to numeric type (int) for classification models
y_temp = pd.to_numeric(y_temp, errors='coerce').astype(int)

# Now, X_temp contains only numeric data, and y_temp is aligned and numeric.
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_temp) # Use X_temp for scaling
X_scaled = pd.DataFrame(X_scaled, columns=X_temp.columns) # Use X_temp.columns for new DataFrame

print(" Features scaled successfully!")
```

```
Scaling features...
 Features scaled successfully!
```

*Figure 25: Screenshot of Scaling Features*

```
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y_temp, test_size=0.2, random_state=42, stratify=y_temp
)

print(f"Train: {X_train.shape[0]:,} | Test: {X_test.shape[0]:,}")
```

```
Train: 24,000 | Test: 6,000
```

*Figure 26: Screenshot of Splitting Data in to Train Test*

### 4.6 Model Training

Model training is a important steps where model learns from the data and prepare for the prediction. The models are fed with the data in this phase, model learns patterns.

Step 5: Model Training

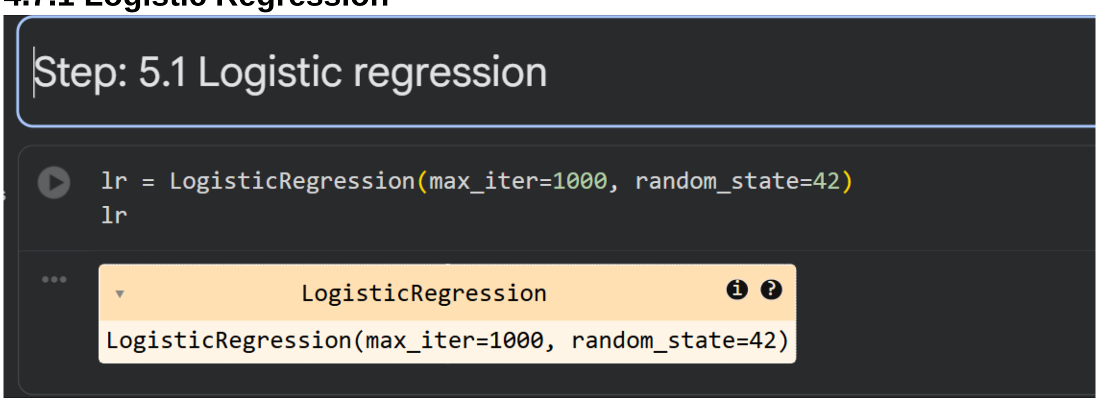
```python
results = {}

def evaluate_model(model, name, X_tr, X_te, y_tr, y_te):
    model.fit(X_tr, y_tr)
    y_pred = model.predict(X_te)
    y_prob = model.predict_proba(X_te)[:, 1]

    acc = accuracy_score(y_te, y_pred)
    prec = precision_score(y_te, y_pred)
    rec = recall_score(y_te, y_pred)
    f1 = f1_score(y_te, y_pred)
    auc = roc_auc_score(y_te, y_prob)

    results[name] = {'Model': model, 'Accuracy': acc, 'Precision': prec,
                     'Recall': rec, 'F1-Score': f1, 'ROC-AUC': auc,
                     'Predictions': y_pred, 'Probabilities': y_prob}

    print(f"Accuracy:  {acc:.4f} ({acc*100:.2f}%)")
    print(f"Precision: {prec:.4f}")
    print(f"Recall:    {rec:.4f}")
    print(f"F1-Score:  {f1:.4f}")
    print(f"ROC-AUC:   {auc:.4f}")

    cm = confusion_matrix(y_te, y_pred)
    print(f"\nConfusion Matrix: TN={cm[0][0]}, FP={cm[0][1]}, FN={cm[1][0]}, TP={cm[1][1]}")
    return model
```

*Figure 27: Screenshot of Model Training*

### 4.7 Model Selection

The models are chosen and used for training on training-dataset. Here for our project, we have chosen logistic regression, random forest and gradient boosting for model training.

#### 4.7.1 Logistic Regression



Step: 5.1 Logistic regression

```python
lr = LogisticRegression(max_iter=1000, random_state=42)
lr
```

```
LogisticRegression
LogisticRegression(max_iter=1000, random_state=42)
```

*Figure 28: Screenshot of Model Selection Logistic Regression*

### 4.7.2 Random Forest

```
Step: 5.2 RANDOM FOREST

rf = RandomForestClassifier(n_estimators=100, random_state=42, n_jobs=-1)
rf

RandomForestClassifier
RandomForestClassifier(n_jobs=-1, random_state=42)
```

*Figure 29: Screenshot of Model Selection Random Forest*

### 4.7.3 Gradient Boosting

```
Step: 5.3 GRADIENT BOOSTING

gb = GradientBoostingClassifier(n_estimators=100, random_state=42)
gb

GradientBoostingClassifier
GradientBoostingClassifier(random_state=42)
```

*Figure 30: Screenshot of Model Selection Gradient Boosting*

## 4.8 Model Evaluation

The trained models are evaluated using classification report. The evaluation matrices for all models are accuracy, precision, recall and f1 score.

### 4.8.1 Evaluation of Logistic Regression

```
[31]
print('Evaluation of Logistic Regression Model')
lr = evaluate_model(lr, "Logistic Regression", X_train, X_test, y_train, y_test)

Evaluation of Logistic Regression Model
Accuracy:  0.8085 (80.85%)
Precision: 0.6918
Recall:    0.2419
F1-Score:  0.3585
ROC-AUC:   0.7078

Confusion Matrix: TN=4530, FP=143, FN=1006, TP=321
```

*Figure 31: Screenshot of Evaluation of Logistic Regression Model*

### 4.8.2 Evaluation of Random Forest

```
[45]
✓ 8s
print('Evaluation of Random Forest Model')
rf = evaluate_model(rf, "Random Forest", X_train, X_test, y_train, y_test)

Evaluation of Random Forest Model
Accuracy:  0.8148 (81.48%)
Precision: 0.6429
Recall:    0.3662
F1-Score:  0.4666
ROC-AUC:   0.7568

Confusion Matrix: TN=4403, FP=270, FN=841, TP=486
```

*Figure 32: Screenshot of Evaluation of Random Forest*

### 4.8.3 Evaluation of Gradient Boosting

```
[46]
✓ 15s
print('Evaluation of Gradient Boosting Model')
gb = evaluate_model(gb, "Gradient Boosting", X_train, X_test, y_train, y_test)

Evaluation of Gradient Boosting Model
Accuracy:  0.8193 (81.93%)
Precision: 0.6685
Recall:    0.3632
F1-Score:  0.4707
ROC-AUC:   0.7790

Confusion Matrix: TN=4434, FP=239, FN=845, TP=482
```

*Figure 33: Screenshot of Evaluation of Gradient Boosting*

## 4.9 Model Comparison

In this section, the model performance is compared and evaluated which model perform the best among three. From the above evaluation metrics, it is hinted that gradient boosting might be the best performing model. To check this, we need to compare the performance or evaluation metrics like accuracy, precision, recall and f1.

**Accuracy:** Proportion of correctly predicted instances out of the total predictions made.

**Precision:** Proportion of correctly predicted positive cases (true positives) out of all cases predicted as positive (true positives + false positives).

**Recall:** Proportion of correctly predicted positive cases out of all true cases.

**F1:** Combination of precision and recall

*Figure 34: Screenshot of Model Comparison*

## Step 6: Model Comparison

[36]

0s

```
comp = pd.DataFrame({
    'Model': list(results.keys()),
    'Accuracy': [results[m]['Accuracy'] for m in results],
    'Precision': [results[m]['Precision'] for m in results],
    'Recall': [results[m]['Recall'] for m in results],
    'F1-Score': [results[m]['F1-Score'] for m in results],
    'ROC-AUC': [results[m]['ROC-AUC'] for m in results]
})

best = comp.loc[comp['Accuracy'].idxmax(), 'Model']
print(f"\n BEST MODEL: {best}")
print(f"Accuracy: {comp['Accuracy'].max()*100:.2f}%\n")
comp
```

```
 BEST MODEL: Gradient Boosting
Accuracy: 81.93%
```

| | Model | Accuracy | Precision | Recall | F1-Score | ROC-AUC |
|---|---|---|---|---|---|---|
| 0 | Logistic Regression | 0.808500 | 0.691810 | 0.241899 | 0.358459 | 0.707757 |
| 1 | Random Forest | 0.814833 | 0.642857 | 0.366240 | 0.466635 | 0.756830 |
| 2 | Gradient Boosting | 0.819333 | 0.668516 | 0.363225 | 0.470703 | 0.779022 |

From the above outcome, it is clear that the best model is gradient boosting with the accuracy of 81.93%.

## 4.10 Model Comparison Visualization by Metrics

Step 7: Model Comparison Visualizations

[37]
```python
#Grah to compare the evaluation metrics for each model
fig, axes = plt.subplots(2, 2, figsize=(15, 10))
fig.suptitle('Model Performance Comparison', fontsize=16, fontweight='bold')

metrics = ['Accuracy', 'Precision', 'Recall', 'F1-Score']
colors = ['#3498db', '#e74c3c', '#2ecc71']

for i, m in enumerate(metrics):
    ax = axes[i//2, i%2]
    bars = ax.bar(comp['Model'], comp[m], color=colors, alpha=0.7, edgecolor='black')
    ax.set_title(m, fontweight='bold')
    ax.set_ylim([0, 1.05])
    ax.grid(axis='y', alpha=0.3)
    for bar in bars:
        h = bar.get_height()
        ax.text(bar.get_x() + bar.get_width()/2., h, f'{h:.3f}', ha='center', va='bottom')

plt.tight_layout()
plt.show()
```

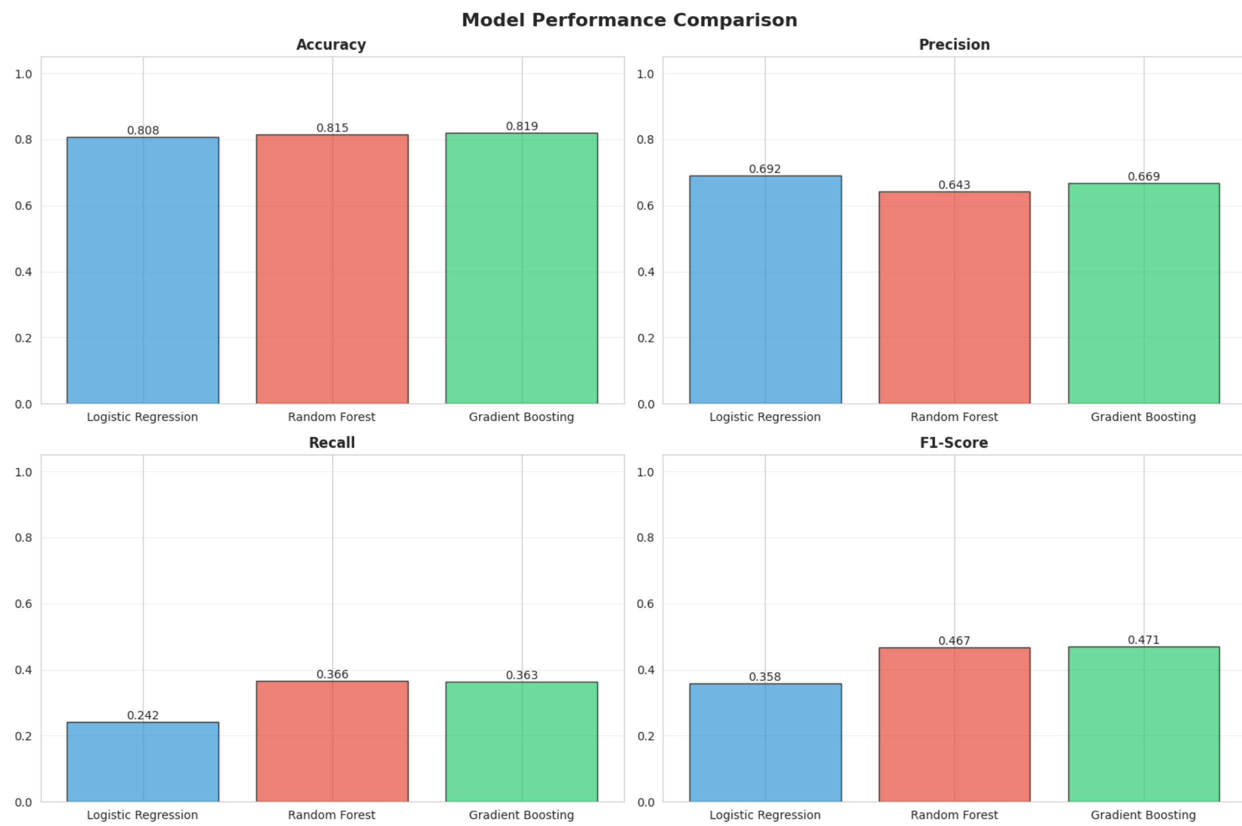



*Figure 35: Screenshot of Model Comparison Performance*

## 4.11 ROC Curve

The ROC curve demonstrates the performance of all three classification models that is Logistic, Rand Forest and Gradient Boosting. They are used to distinguish the default and non-default credit card clients. Each curve plots the True Positive rate against False Positive rate. It shows that the model balances true positive and false positive correctly.

```
[38]
# ROC curve for 3 models
plt.figure(figsize=(10, 8))
for name in results:
    fpr, tpr, _ = roc_curve(y_test, results[name]['Probabilities'])
    plt.plot(fpr, tpr, label=f"{name} (AUC={results[name]['ROC-AUC']:.3f})", linewidth=2)

plt.plot([0, 1], [0, 1], 'k--', label='Random')
plt.xlabel('False Positive Rate', fontsize=12)
plt.ylabel('True Positive Rate', fontsize=12)
plt.title('ROC Curves', fontsize=14, fontweight='bold')
plt.legend()
plt.grid(alpha=0.3)
plt.show()
```

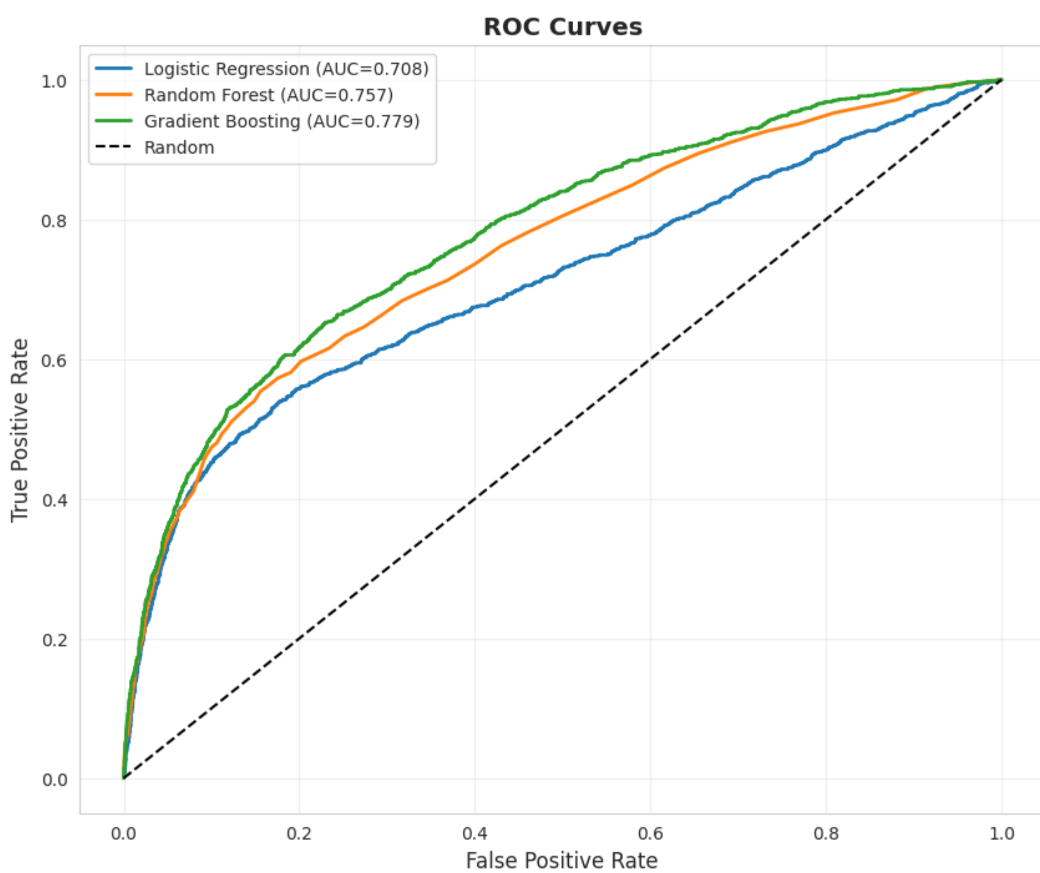



*Figure 36: Screenshot of ROC Curve of Models*

## 4.12 Confusion Matrix of Models

```
[39]
# Confusion matrix of all models
fig, axes = plt.subplots(1, 3, figsize=(15, 4))
fig.suptitle('Confusion Matrices', fontsize=14, fontweight='bold')

for i, name in enumerate(results):
    cm = confusion_matrix(y_test, results[name]['Predictions'])
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', ax=axes[i], cbar=False)
    axes[i].set_title(name)
    axes[i].set_xlabel('Predicted')
    axes[i].set_ylabel('Actual')

plt.tight_layout()
plt.show()
```

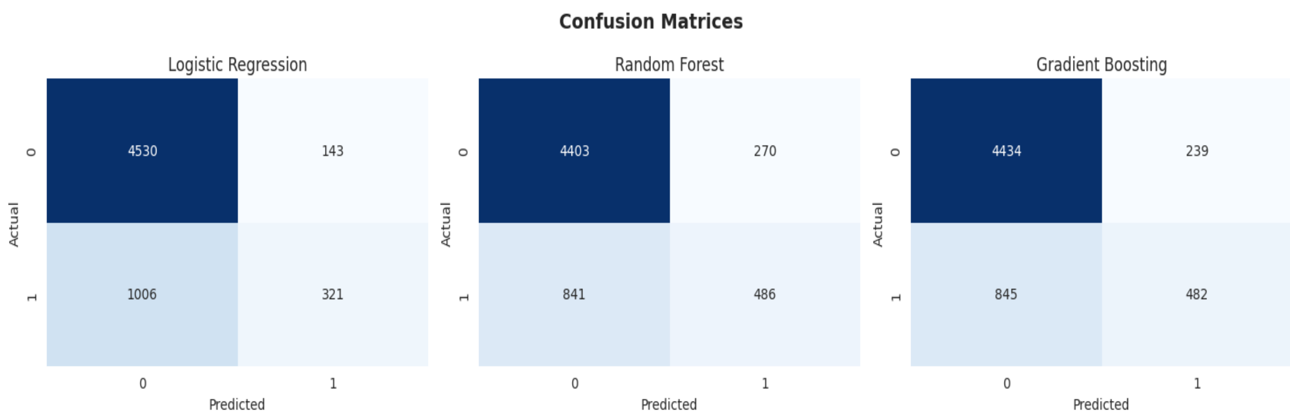



*Figure 37: Screenshot of Confusion matrix of all Models*

### 4.13 Feature Importance

Feature importance is one of the significant step to check after model prediction as it shows us which feature influenced most for the prediction. In the system like credit card default prediction, knowing which feature contributed more to the model or influence most helps businesses and financial institution to understand the credit risk factor.

Step 8: Feature Importance

[41]
0s

```
feat_imp = pd.DataFrame({
    'Feature': X.columns,
    'Importance': rf.feature_importances_
}).sort_values('Importance', ascending=False)

print("Top 15 Features:")
feat_imp.head(15)
```

```
Top 15 Features:
```

| | Feature | Importance |
|---|---|---|
| 6 | X6 | 0.093286 |
| 0 | Unnamed: 0 | 0.071426 |
| 5 | X5 | 0.056722 |
| 12 | X12 | 0.055043 |
| 1 | X1 | 0.051537 |
| 13 | X13 | 0.049945 |
| 18 | X18 | 0.047794 |
| 14 | X14 | 0.047522 |

*Figure 38: Screenshot of Feature Importance*

```
# Graph for most important features
plt.figure(figsize=(10, 8))
top = feat_imp.head(15)
plt.barh(range(len(top)), top['Importance'], color='skyblue', edgecolor='black')
plt.yticks(range(len(top)), top['Feature'])
plt.xlabel('Importance')
plt.title('Top 15 Feature Importance', fontsize=14, fontweight='bold')
plt.gca().invert_yaxis()
plt.grid(axis='x', alpha=0.3)
plt.tight_layout()
plt.show()
```

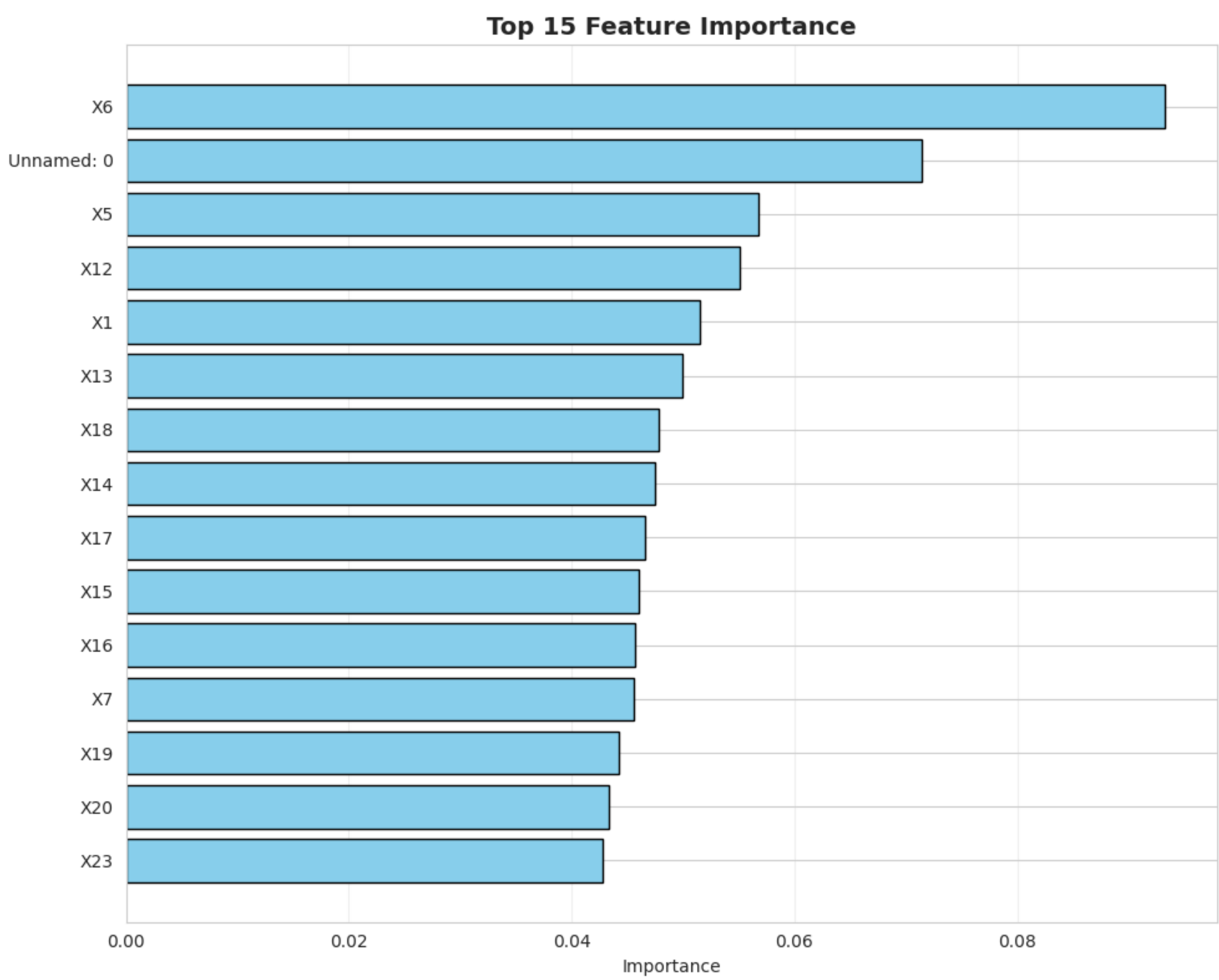



*Figure 39: Screenshot of Top 15 Important Features*

# 5. Conclusion

## 5.1 Analysis of the work done

The project serves as solution to the problem of credit default using the concept of machine learning and artificial intelligence algorithm. The coursework includes the development of a system with at least of 3 algorithms to compare their analysis. During the development of the project, the concepts like supervised learning, models and other required things are studied and implemented properly. The research papers journals and articles have also studied that helps us understand the work done in this area.

Overall, the system is eligible for predicting the best outcome as it compares the prediction of other models. The concept of ensembling the predicted data is also learned during the coursework. The methodology followed a structure ML pipeline, including dataset loading, processing, training, evaluation ensuring reliable results of the dataset. The project solves the problem of financial institution and banks by reducing financial losses and identifying the risky clients causing the debt to minimize.

## 5.2 Solution to Real World Problem

The project addressed the real-world problems by tackling one of the most critical challenges in the financial sectors. The system now can correctly identify and predict the default and non-default to client based on their age, behavior, repayment history and other important features. This enables financial sectors to make informed and data driven decision such as adjusting credit limits, offering tailored repayment plans, or tightening approval criteria. Not only in banking and finance sectors but the solution also contribute to fairer lending practices by using data-driven insights rather than relying on traditional approach. It helps people and business understand and identify the loyal clients to whom can be payment time and credit limit can be extended while the people unlike to pay can be tighten.

### 5.3 Future Works

The future works for the system will most be integrating it with the banking system or financial companies to work in real world. Expanding more features and exploring neural networks for high performance, accepting multiple requests at a time. The model is currently showing best result with gradient boosting, an ensemble method as a future work we can test it with more advanced ensemble methods like XGboost, LightGBM, or CatBoost, which often outperformed the Gradient Boosting in both speed and accuracy.

In addition to model experimentation, a future-research may incorporate explainable artificial intelligence (XAI) tools, e.g. SHAP or LIME, to be able to give a better understanding of why the model predicts default of some clients. This would render the solution more viable to the financial institutions in need of accountability in decision-making. Lastly, scaling the dataset with other behavioral or transactional characteristics (e.g. spending habits, how frequently they pay back) or to other financial datasets would probe the generalizability of your algorithm and enhance its scale to the real world.

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